Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, September - 2021
DESIGN PATTERNS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
- - -

1. Draw a diagram to show relationships among design patterns? How to select a design pattern? How to use a design pattern? [15]

2. Explain the following with respect to LEXI's design:
   a) Establishing the user interface
   b) Supporting multiple Window systems. [7+8]

3. Discuss about creational patterns. [15]

4.a) Explain about sample code and uses of Bridge pattern.
   b) Discuss about implementation and related patterns of Composite pattern. [8+7]

5. Write and explain sample codes of Decorator and Flyweight patterns in detail. [15]

6. Discuss in detail about Chain of Responsibility pattern. [15]

7. Describe applicability, collaborations and sample code of Memento and Strategy. [15]

8. Explain the following:
   a) Alexander's pattern languages
   b) History of design patterns
   c) A common design vocabulary. [5+5+5]

--ooOoo--